Random Number Generators
Algorithms for Gaussian Samples

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Based on Section 2 of: D. B. Thomas, W. Luk, P. H. W. Leong, J. D. Villasenor, *Gaussian Random Number Generators*, ACM Computing Surveys (2007).

Scope: Selected Section 2 Methods

- Focus on three generators from Section 2:
  - Central Limit Theorem (CLT) sum
  - Piecewise linear (triangles) approximation
  - Monty Python packing method
- For each: algorithm idea, distribution graph, QQ plot, and complexity.

Standard Normal: PDF and CDF

Probability density function (PDF)

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Cumulative distribution function (CDF)

$$\Phi(x) = \int_{-\infty}^{x} \phi(t)\, dt = \tfrac{1}{2}\Big(1 + \mathrm{erf}(x/\sqrt{2})\Big).$$

- No closed form for Φ or Φ^{-1} $\Rightarrow$ numerical/approximate evaluation.

Approximate vs. Exact (This Talk)

Approximate methods

- CLT sum (finite K)
- Piecewise linear triangles
- Fast/simple, but tails deviate

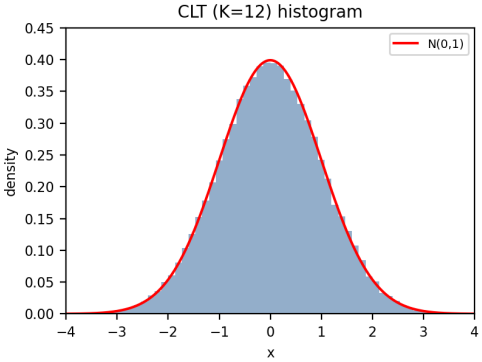
Exact transform

- Monty Python packing
- Exact if tail sampler is exact
- Fast path dominates most samples

Central Limit Theorem (CLT) Method

- Generate $U_i \sim \mathcal{U}(0,1)$ for $i = 1, \ldots, K$ and sum $S = \sum U_i$.
- Standardize to unit variance:
$$Z = \frac{S - K/2}{\sqrt{K/12}}.$$
- For $K = 12$, $\sqrt{K/12} = 1$, so $Z = S - 6$ is already $\mathcal{N}(0,1)$-scaled.
- Exact for the Irwin–Hall distribution, but only **approximate** to Gaussian; tails are too light for finite K.

CLT Graph: Histogram vs. Normal

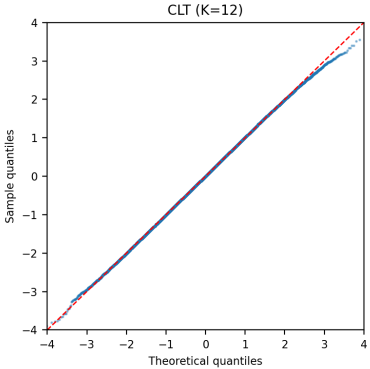


Histogram of CLT samples (K=12) with $\mathcal{N}(0,1)$ overlay; the center matches well while tails are thin.

QQ Plots: How to Read Them

- X-axis: theoretical quantiles of $\mathcal{N}(0,1)$; Y-axis: sample quantiles.
- Points near the 45° line $\Rightarrow$ good distributional match.
- Inward bend at extremes $\Rightarrow$ light tails; outward bend $\Rightarrow$ heavy tails.
- Kinks/steps suggest piecewise approximations; scatter reflects sampling noise.

CLT QQ Plot



Inward bending in the tails confirms the thin-tail bias of finite K sums.

Piecewise Linear (Triangles) Method

- Approximate $\phi(x)$ on $[0, x_{\max}]$ by linear segments (triangles), then mirror.
- Choose triangle i with probability proportional to its area (alias table).
- Sample within the triangle using two uniforms; apply a random sign.
- Accuracy improves with more segments and larger $x_{\max}$, but cost grows.

Triangles: Basic Idea

- Target: $Z \sim \mathcal{N}(0,1)$ with $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.
- Approximate with a mixture of triangular densities:

$$\phi(x) \approx g(x) = \sum_{i=1}^{k} q_i \, t_i(x),$$

- where $q_i \geq 0$ and $\sum_{i=1}^{k} q_i = 1$.

Triangles: Triangular Component Density

$$t_i(x) = \begin{cases} \frac{1}{w^2}(w - |x - c_i|), & |x - c_i| \leq w, \\ 0, & \text{otherwise.} \end{cases}$$

- Each triangle has width $2w$ and center c_i.
- Area check: $\frac{1}{2}(2w)\left(\frac{1}{w}\right) = 1$.

Triangles: Placement and Weights

- Centers equally spaced:

$$c_i = w\left(\frac{k+1}{2} - i\right), \quad i = 1, \ldots, k.$$

- Neighboring triangles overlap, yielding a piecewise-linear bell shape.
- Weight choice to mimic Gaussian mass:

$$q_i \propto \phi(c_i), \qquad q_i = \frac{\phi(c_i)}{\sum_{j=1}^{k} \phi(c_j)}.$$

Triangles: Advantages and Limitations

Advantages

- Only adds and multiplies.
- Hardware-friendly.
- Good central accuracy.

Limitations

- Approximate, not exact.
- Tails less accurate.
- More uniforms as k grows.

Triangles Approximation (Fig. 1)

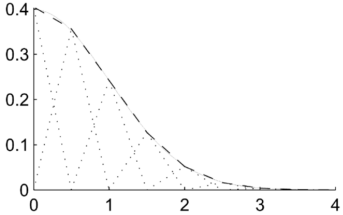
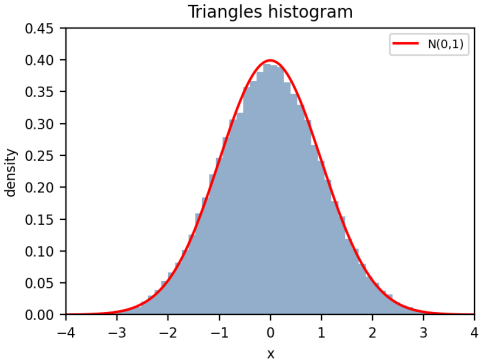


Fig. 1. Approximation to the Gaussian distribution composed of multiple triangle distributions.

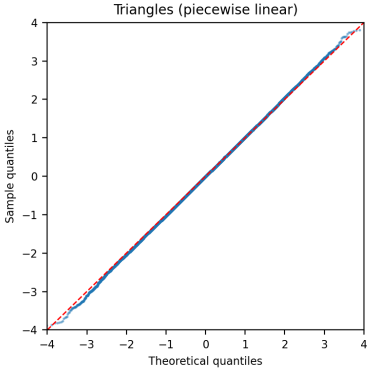
Fig. 1 from Thomas et al. (2007): Gaussian approximation using multiple triangle distributions.

Triangles Graph: Histogram vs. Normal



Histogram shows close central fit with small deviations near the tails and segment boundaries.

Triangles QQ Plot



Small kinks reflect the piecewise-linear approximation; tails deviate from the line.

Monty Python Method (Packing)

- Fold the Gaussian PDF into a rectangle of width b and height $1/(2b)$.
- Sample a uniform point (x, y) in the rectangle and identify region:
  - Areas A/B: return x directly (fast path).
  - Area C: affine map back; accept if under $\phi(x)$.
  - Area D: draw from the tail $|x| > b$ with a dedicated tail sampler.
- Parameter b trades tail frequency vs overlap (e.g., $b \approx 2.29$ gives $\sim$2.2% tail calls).

Monty Python: Symmetry Reduction

- Sample the positive half, then attach a random sign.

$$Y = |Z|, \qquad Z = SY, \quad P(S = 1) = P(S = -1) = \tfrac{1}{2}.$$

Monty Python: Rectangle Construction

Choose $b > 0$ and define:

$$0 \le x \le b, \qquad 0 \le y \le \frac{1}{2b}.$$

Area:

$$b \cdot \frac{1}{2b} = \frac{1}{2}.$$

Sample:

$$X \sim \mathcal{U}(0, b), \qquad Y \sim \mathcal{U}\left(0, \frac{1}{2b}\right).$$

Monty Python: Partition into Regions

Split the rectangle into:

$$A,\ B,\ C',\ D'.$$

- A: always accept ($\phi(x) \geq 1/(2b)$).
- B: accept if $y < \phi(x)$.
- C': affine map into C, then re-test.
- D': tail region $|x| > b$ handled separately.

Monty Python: Region A and B

Define a by:

$$\phi(a) = \frac{1}{2b}.$$

If $x < a$, then $y < \phi(x)$ automatically (accept). Otherwise in B, accept if:

$$y < \phi(x).$$

Monty Python: Region C' and D'

- C': map $(x, y) \mapsto f_C(x, y)$ into C; accept if under $\phi(x)$.
- The transform preserves area, so probabilities remain correct.
- D': corresponds to $|x| > b$; use a separate exact tail sampler.

Monty Python: Why It Is Exact

1. Uniform rectangle sampling is exact.
2. Accepted points under ϕ match the Gaussian density.
3. Area-preserving transforms keep probabilities correct.
4. Tail sampler is exact.
5. Random sign restores symmetry.

Monty Python Packing (Fig. 2)

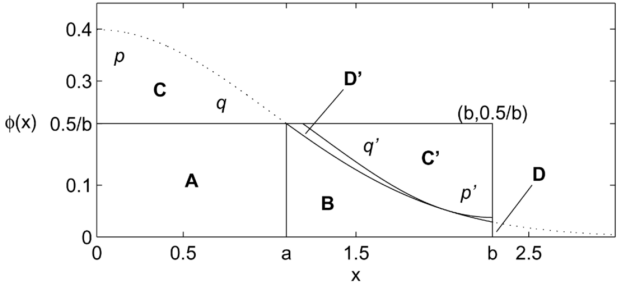
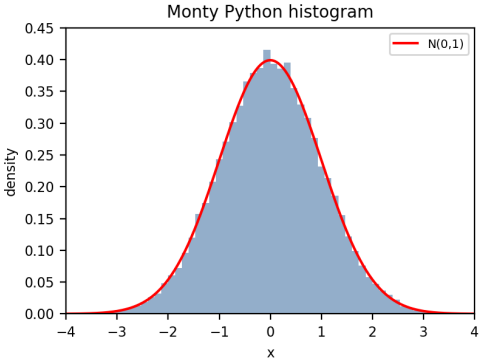


Fig. 2. Packing of the Gaussian distribution into a rectangular area using the Monty Python method.

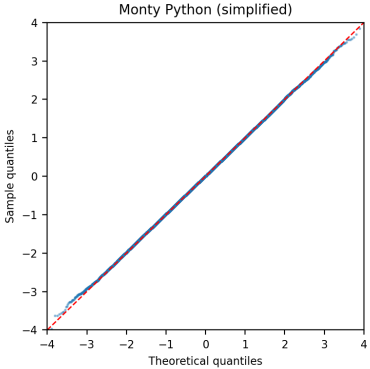
Fig. 2 from Thomas et al. (2007): Packing of the Gaussian distribution into a rectangle.

Monty Python Graph: Histogram vs. Normal



Histogram closely matches $\mathcal{N}(0,1)$; differences mostly driven by tail routine.

Monty Python QQ Plot



Points follow the 45° line across most quantiles, indicating near-exact behavior.

Complexity Comparison

Method	Avg cost (uniforms / ops)	Notes
CLT sum (K)	K uniforms $+$ $K-1$ adds (O(K))	Approximate; thin tails unless K is large.
Triangles (piecewise)	2–3 uniforms $+$ alias/table $+$ mult/add	Approximate; quality depends on segments and $x_{\max}$.
Monty Python	2–3 uniforms fast path; occasional $\phi(x)$/tail	Exact if tail sampler exact; efficiency set by b.

Takeaways

- CLT is simplest but has visibly light tails unless K is large.
- Triangles trade accuracy for speed; more segments and larger $x_{\max}$ improve fit.
- Monty Python is near-exact with a fast path; performance depends on tail frequency.

Reference

D. B. Thomas, W. Luk, P. H. W. Leong, and J. D. Villasenor.
Gaussian Random Number Generators.
ACM Computing Surveys, 39(4), Article 11, 2007.
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